Genetic Algorithm

Pseudo code

```
def fitness(x):
    return x

population = generate random population

population_size = int(input("Enter population size"))
mutation_rate = int(input("Enter the mutation rate"))
crossover_rate = int(input("Enter rate of crossover"))
generation = int(input("Enter the generation"))

for generation in (generation):

    fitness value = []
    for chromosome in population:
        x = binary to decimal (chromosome)

    fitness value.append(x)
        parents = select best (population fitness value)
        offspring []

while length (offspring) < population size
    parent 1 = select parent (parents)
    parent 2 = select parent (parents)
```

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```
if random() < cross rate
        child = crossover [parent1, parent2]
else:
    child 1 = parent1
    child 2 = parent2
    offspring append child 1
    offspring append child 2

for child in offspring
        mutate (child 1, child 2)
    population = offspring
    best = find best (population)
    x = binary to decimal (best)

print (x)
print 'fitness (x)
end
```

M/c learning.

Machine learning using genetic algorithm for feature selection

Application:

Title of the paper → "Android malware detection using GA-based optimized feature selection & machine learning"

This paper uses GA because feature selection is an optimization problem. when there are many features, there can be a very large number of possible feature combination. GA efficiently searches these combination using selection, crossover & mutation. The resulting feature subset can then be used by a machine-learning classifier."

Code summary

This code implements a genetic algorithm for feature selection in machine learning. A binary chromosome represents the selected & unselected features. A SVM classifier is used to evaluate the fitness of each chromosome based on classification accuracy. The algo repeatedly performs selection, crossover & mutation to find a feature subset with better performance

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Output Summary

The program displays the best fitness accuracy for each generation. At the end, it displays the best chromosome, the number of selected features, & the best classification accuracy obtained. The output shows how GA improves the feature selection process by evaluating successive generations.

Proceed.

Coding is completed

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